

MAT3032 ADVANCED ALGEBRA: UNASSESSED COURSEWORK 1

The answers are to be entered in Möbius (MapleTA) using the links in Course Materials > Unassessed Coursework > Coursework 1 on SurreyLearn. You are advised to work through the questions on paper first. There are separate assignments/links for Questions 1, 2, 3 and 4/5. After submitting each part, select 'view details' to check your answers. Please repeat the assignments until you are happy with your marks, and complete them by **16:00 on Friday 18th October 2024**.

1. This question is about groups of order 8. Construct a Cayley table for each of the groups described below. State whether each group is abelian or non-abelian. In each group, state the number of elements of order 2, i.e. non-identity elements x such that x^2 is the identity.

- (a) $\mathbb{Z}_8 = \{0, 1, 2, 3, 4, 5, 6, 7\}$ under addition modulo 8, e.g. $4 +_8 7 = 3$.
- (b) $\mathbb{Z}_2 \times \mathbb{Z}_4 = \{(0, 0), (0, 1), (0, 2), (0, 3), (1, 0), (1, 1), (1, 2), (1, 3)\}$ with the operation $(a, b) + (c, d) = (a +_2 c, b +_4 d)$, e.g. $(1, 2) + (1, 3) = (0, 1)$.
- (c) $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 = \{(a, b, c) : a, b, c \in \{0, 1\}\}$ with the operation $(a, b, c) + (d, e, f) = (a +_2 d, b +_2 e, c +_2 f)$, e.g. $(1, 1, 0) + (0, 1, 1) = (1, 0, 1)$.
- (d) $\left\{ \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} : a, b, c \in \{0, 1\} \right\}$ under matrix multiplication (modulo 2).

This group is generated by $\{A, B\}$ where $A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$,

the elements being $I, A, A^2, A^3, B, AB, A^2B, A^3B$.

Is the group isomorphic to the dihedral group D_4 ?

- (e) $\{1, -1, i, -i, j, -j, k, -k\}$ under multiplication (which is associative) where the quantities i, j, k satisfy the equations $i^2 = j^2 = k^2 = -1$, $ij = k$, $jk = i$, $ki = j$, and 1 satisfies the usual conditions for a multiplicative identity, so $1i = i$, $j(-1) = -j$, $(-1)^2 = 1$, $ji = j(jk) = j^2k = -k$, etc. Is this group isomorphic to D_4 ?

It will be shown in the module that, up to isomorphism, the five groups in Question 1 are the only groups of order 8.

A group isomorphism preserves the order of each element. Thus if two groups have different numbers of elements of a particular order, they cannot be isomorphic. Also, an abelian group cannot be isomorphic to a non-abelian group.

The group of units of $(\mathbb{Z}_{10}, \times_{10})$ is $U_{10} = (\{1, 3, 7, 9\}, \times_{10})$. We can decide which group of order 4 this is without making a complete Cayley table, by finding the number of elements of order 2. To do this, simply square each element:

x	1	3	7	9
x^2	1	9	9	1

(Note that, working modulo 10, 9^2 can be evaluated as $(-1)^2$.)

There is just one element, namely 9, that has order 2 in U_{10} . This identifies the group as the cyclic group C_4 , not the Klein 4-group V which has three elements of order 2.

In the next question we apply this approach to some 8-element groups of units.

[TURN OVER]

2. (a) Use Euler's φ function to show that the groups of units U_{15} , U_{16} , U_{20} , U_{24} and U_{30} all have order 8.

(b) Find the number of elements of order 2 in each case. Hence state which of the groups in Question 1 is isomorphic to each of these groups of units.

3. Let p be prime, so $\mathbb{Z}_p = \{0, 1, \dots, p-1\}$ is a field, and let G be the set of all matrices of the form $\begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix}$ where a, b, c can be any elements of $\mathbb{Z}_p$.

(a) Show that G , under matrix multiplication (modulo p), is a subgroup of $GL(3, \mathbb{Z}_p)$. Is G abelian? Find the order of G .

(b) A map $\psi : G \rightarrow (\mathbb{Z}_p, +_p)$ is defined by $\psi : \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \mapsto a +_p c$. Show that ψ is a group homomorphism. Find its kernel and image, and state their orders.

(c) Give a reason why the left and right cosets of $\text{Ker } \psi$ in G are the same. Find a condition for two elements of G to be in the same coset of $\text{Ker } \psi$.

(d) Find a number k such that there must be at least k non-isomorphic groups of order $|G|$. How many of these groups are cyclic? (Recall that $C_m \times C_n \cong C_{mn}$ if and only if m and n are coprime.)

(e) Find how many groups of order $|G|$ there are when $p = 3$, using

http://www.math.ucsd.edu/~atparris/small_groups.html

4. Find the permutation $\tau\sigma\tau^{-1}$ in S_5 , in each of the cases:

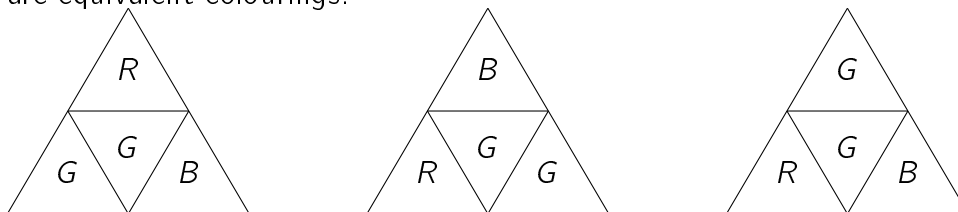
(a) $\sigma = (1 \ 4 \ 2 \ 3 \ 5)$, $\tau = (2 \ 4 \ 5)$,

(b) $\sigma = (1 \ 4 \ 2)(3 \ 5)$, $\tau = (2 \ 4 \ 5)$, (c) $\sigma = (1 \ 2 \ 3 \ 4)$, $\tau = (4 \ 5)$.

In each case give your answer as a cycle or a product of disjoint cycles, and find the smallest value of k for which it can be expressed as a product of k transpositions. Hence state whether $\tau\sigma\tau^{-1}$ is an element of the alternating group A_5 .

5. An equilateral triangle is cut out of transparent material. Lines are drawn to divide it into four congruent triangles, as shown below. Each section is coloured red, blue or green. Each of these colours may be used any number of times.

Colourings are regarded as the same if one can be obtained from another by rotating the triangle in the plane and/or turning it over. For example, the following are equivalent colourings:



Find the numbers of colourings in which 1, 2 and 3 different colours are used. Hence find the total number of distinguishable ways of colouring the figure.

(In Chapter 2 we shall see an easier way of solving this type of problem, using group theory.)